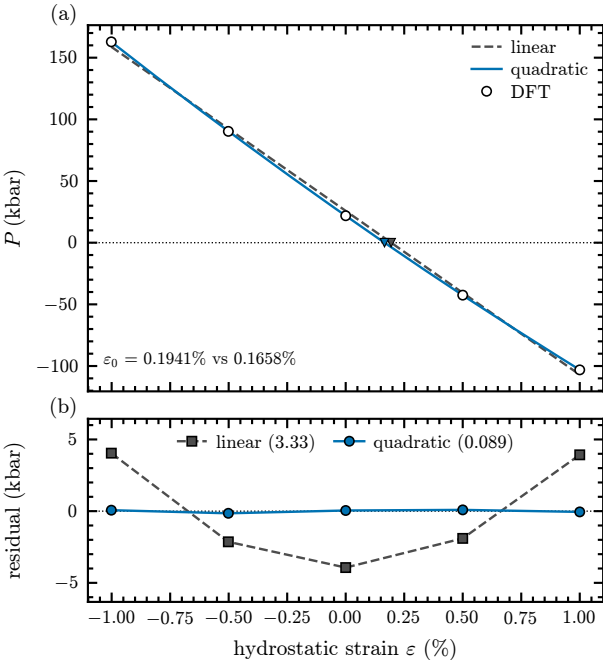




Bulk diamond equation of state at 90/720 Ry, and the bias in a linear $P(\varepsilon)$ fit. (a) $P(\varepsilon)$ is visibly curved over $\pm 1\%$ strain, so the straight-line fit misses the zero-pressure crossing: the two markers on $P = 0$ differ by 0.0283% in strain, which is 0.0010 Å in a_0 . (b) The residuals make the bias unmistakable: the linear fit leaves 3.33 kbar rms with obvious systematic curvature, 37 times the 0.089 kbar of the quadratic. The production reference is neither of the fits drawn here but a 3rd-order Birch-Murnaghan fit to $E(V)$ over the same series, giving $a_0 = 3.572997$ Å and $B = 433.66$ GPa; the quadratic $P(\varepsilon)$ fit shown in (a) is its cross-check, not its source.